Making the Hadamard test scale

36× fewer gates with AQC, a phase trap that would have broken it, and a way to certify the result from its own garbage

Team 8 — Garbage Collectors

TBD-NAME TBD-NAME TBD-NAME

Qiskit Hackathon 2026 — Topic 5

after Faehrmann, Eisert & Kueng, *PRL* **135**, 150603 (2025)

14 August 2026

Outline

The question

The instrument

Against the baselines that already exist

AQC: the compressor and its certificate

On real hardware

What we claim

A tag like [R042] is not decoration — it is a **row number** in our public evidence ledger, `RESULTS.md`: one row per reported number, each with the exact command that produced it. Full repo via the QR code on the last slide. Where a baseline beats us, the slide says so.

The result, up front

- **AQC makes the Hadamard test scale.** Its controlled evolution is the bottleneck: the standard exact block costs $\sim 4^{n+1}$ two-qubit gates. We measured the crossover at $n = 4$ and $36\times$ **fewer gates by** $n = 7$ ($16{,}812 \to 468$), at state infidelity 3×10^{-4}. [R046]
- **But shipping AQC into a Hadamard test silently breaks it.** The compiler optimises *state fidelity*, which is blind to global phase — and a Hadamard test is an interferometer that measures exactly that. A converged compression returns χ wrong by **2.3–3.0 radians**, an error *as large as the signal*. [R046]
- **One ancilla gate fixes it.** $\theta = \arg\langle U\psi | W\psi \rangle$ is free at compile time; a $P(-\theta)$ on the ancilla cancels it exactly. $|\Delta\chi|$: **1.33** $\to$ **0.008**. [R046]
- **And Track B certifies the compressed circuit from shots** — the check that still works where classical simulation does not. [R042, R044]

Reported against us, in full, later: on today's hardware at n=2 a 2-gate Loschmidt echo still beats our 136-gate Track B arm by $27\times$ (slide 18). [R044]

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You compiled U into something cheaper. Is it still right?

Every practical quantum simulation replaces the target evolution $U(t)$ with a cheaper approximation W — a Trotter product formula, a transpiler setting, or an **AQC-compressed ansatz**.

The question that decides whether the result means anything:

how wrong is W, and wrong in *what*?

Classically you answer it by simulating both. That is exactly the thing you cannot do at the sizes where you need the answer.

Track B answers it *on the device*: put W on one ancilla branch and U on the other, and the interference is the overlap of two dynamics.

W = what you can afford to run

U = what you meant to run

$$\chi_{AB}(t) = \langle\psi|W^\dagger U|\psi\rangle$$

measured, not simulated

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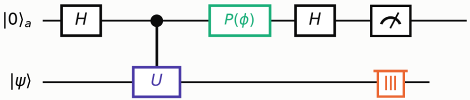
Anti-control: how the Hadamard test is extended

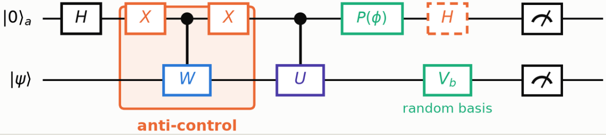
Standard Hadamard test



$\langle Z_a\rangle_\phi = \mathrm{Re}\left[e^{i\phi}\langle\psi|U|\psi\rangle\right]$

the system register is discarded

Track B: the anti-controlled (extended) Hadamard test



$\langle Z_a\rangle_\phi = \mathrm{Re}\left[e^{i\phi}\langle\psi|W^\dagger U|\psi\rangle\right]$

the register is now a classical shadow

dashed H: delete it for arm B
(X-basis readout $\Rightarrow$ the difference)

Why the two X gates:

$X_a(|0\rangle\langle 0|\otimes I + |1\rangle\langle 1|\otimes W)X_a = |1\rangle\langle 1|\otimes I + |0\rangle\langle 0|\otimes W$

Conjugating by X swaps which ancilla state fires the gate, so W acts on the $|0\rangle$ branch and U on the $|1\rangle$ branch. Nothing else changes.

What the ancilla then sees:

$\frac{1}{\sqrt{2}}\left(|0\rangle\, W|\psi\rangle + e^{i\phi}|1\rangle\, U|\psi\rangle\right)$

Two dynamics interfering on one ancilla. $W = I$ gives back the standard test exactly, so this is a strict generalisation.

The only structural change is the X–cW–X sandwich: it makes W fire on the $|0\rangle$ branch, so the ancilla interferes **two dynamics** rather than one against the identity. Everything downstream is unchanged. Algebra in appendix A1–A2.

Track B: anti-controlled Hadamard test

Replace controlled-U with "W when the ancilla is $|0\rangle$, U when it is $|1\rangle$". Then, with the *same* quadrature convention as the ordinary test,

$$\langle Z_a\rangle_\phi = \mathrm{Re}\left[e^{i\phi}\langle\psi|W^\dagger U|\psi\rangle\right], \qquad \phi = 0 \to \mathrm{Re}\,\chi_{AB}, \quad \phi = -\tfrac{\pi}{2} \to \mathrm{Im}\,\chi_{AB}$$

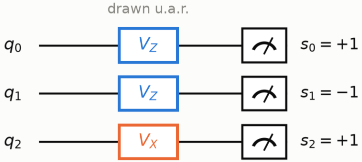
Setting $W = \mathbb{1}$ recovers the ordinary Hadamard test exactly, so this is a strict generalisation — which also means it needs *its own* validation, not inherited trust.

statevector identity, frozen 3-site (3 times $\times$ 2 reps $\times$ 2 quadratures)	8.99×10^{-15} [R034]
statevector identity, 2-site (4 times $\times$ 2 reps $\times$ 2 quadratures)	4.56×10^{-15} [R042]
$\chi_{AB}(0) = 1$ exactly, as it must be (both branches are the identity)	**yes**

A bug this caught. Our first reference took the $|1\rangle$-control block as the *upper half* of the gate matrix. The notebook builds the gate with the ancilla as its *lowest* bit, so the block is the odd sublattice. The wrong version passes at $t = 0$ — where both branches are the identity — and fails everywhere else. It was caught before any QPU time was spent.

First, what a classical shadow is

1. Each shot draws its own random basis



$b_j \sim \mathrm{Unif}\{X, Y, Z\}$, independently on every qubit and every shot

All that is stored per shot: $(\, b_0 b_1 b_2,\;\; s_0 s_1 s_2 \,)$

$$\hat{\rho} = \bigotimes_j \Big(3\, V_{b_j}^\dagger |s_j\rangle\langle s_j| V_{b_j} - \mathbb{1} \Big), \qquad \mathbb{E}[\hat{\rho}] = \rho$$

one unbiased snapshot — meaningless alone, exact in expectation once averaged

2. One record, every Pauli at once

$$\hat{P} = 3^{\,w(P)} \prod_{j \in \mathrm{supp}(P)} s_j \; \mathbf{1}[b_j = P_j]$$

shot	**basis** b	**outcome** s	$\widehat{Z_0 Z_1}$	$\widehat{X_0 X_1}$
1	**Z Z X**	$+ - +$	-9	0
2	**X X Z**	$+ + -$	0	$+9$
3	**Z Y Y**	$- + +$	0	0
4	**Z Z Z**	$+ + +$	$+9$	0
5	**Y X Z**	$- - +$	0	0

$w = 2$: a shot lands with probability 3^{-2} and the 3^2 weight cancels it exactly, so $\mathbb{E}[\hat{P}] = \mathrm{Tr}[P\rho]$

q_2's basis never enters $\widehat{Z_0 Z_1}$: only $\mathrm{supp}(P)$ must match

$\Rightarrow$ 3^{-w}, not 3^{-n}. The price is $\mathbb{E}[\hat{P}^2] = 3^{\,w}$.

In Track B this is applied to the *system* register — the one a standard Hadamard test throws away. It costs **no new circuits**, and it makes one record set an unbiased estimator for **every** Pauli at once.

(Schematic uses 3 qubits; our Track B instance has 2.) [Huang, Kueng & Preskill, *Nat. Phys.* **16**, 1050 (2020); proof in A4]

What the shadows add: delete the final Hadamard

Before the final Hadamard the state is $\big(|0\rangle W|\psi\rangle + e^{i\phi}|1\rangle U|\psi\rangle\big)/\sqrt{2}$. Tracing the ancilla out of the *un-Hadamarded* circuit gives two different things at once:

$$\mathrm{tr}_a\big[\rho_{\text{out}}\big] = \tfrac{1}{2}\big(W\rho W^\dagger + U\rho U^\dagger\big), \qquad \mathrm{tr}_a\big[(Z\otimes \mathbb{1})\rho_{\text{out}}\big] = \tfrac{1}{2}\big(W\rho W^\dagger - U\rho U^\dagger\big)$$

Both are *independent of* ϕ, so one quadrature suffices. Add and subtract:

$\langle O\rangle_W$ and $\langle O\rangle_U$ separately, for every Pauli O, from one family of circuits.

That is the *per-observable disagreement profile* — the part of Track B the scaffold flags and that our own R034 explicitly did **not** deliver. Validated to 4.44×10^{-15} across 4 times $\times$ 6 observables before submission. [R042]

The ancilla says *how much* W is wrong. The garbage says *which observable*.

The question

The instrument

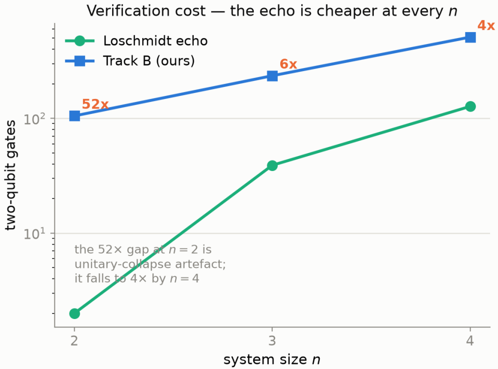
Against the baselines that already exist

AQC: the compressor and its certificate

On real hardware

What we claim

We are not the only way to do this — and not the cheapest



What each method actually returns

	classical fidelity	Loschmidt echo	Hilbert-Schmidt	Track B ancilla only	Track B + shadows
gives a number	yes	yes	yes	yes	yes
runs on a QPU	--	yes	yes	yes	yes
keeps the phase	--	--	--	yes	yes
says WHICH observable broke	--	--	--	--	yes

Loschmidt echo: prepare, apply U, apply $W^\dagger$, un-prepare, read the all-zeros probability.

Hilbert–Schmidt test (Khatri *et al.*, *Quantum* **3**, 140 (2019)): Bell pairs on $2n$ qubits give the *state-independent* $|\mathrm{Tr}[W^\dagger U]/d|^2$. Both verified against exact values before being used as baselines.

[R043]

The honest scoreboard

	qubits	2q gates	phase?	state-indep.?	which observable?
classical fidelity (AQC's own score)	—	—	n/a	yes	no
Loschmidt echo	n	2	no	no	no
Hilbert–Schmidt test	$2n$	16	no	yes	no
Track B, ancilla only	$n+1$	136	yes	no	no
Track B + shadows (ours)	$n+1$	136	yes	no	yes

- **The echo is cheaper than us at every n we measured** — 6.0× at $n=3$ and 4.0× at $n=4$. If all you want is a scalar "how close is W to U", *use the echo*. [R043]
- The 52× gap at $n=2$ is a *unitary-collapse artefact* (any product of 2-qubit unitaries fuses into one). We quote 4–6×, never 52×.
- The HST answers a *stronger* question than either overlap, and costs $2n$ qubits for it.

Our arm has to be justified by what it returns, never by being cheaper.

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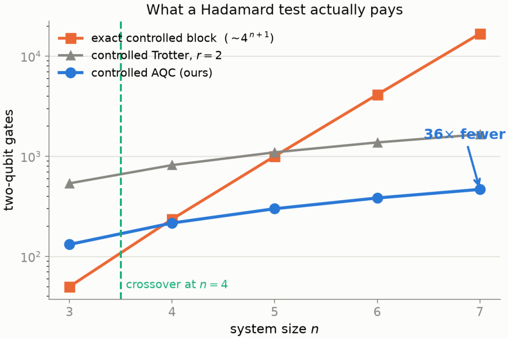
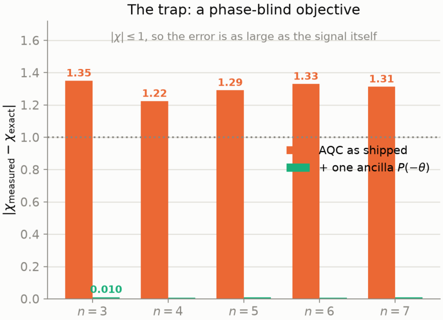
AQC makes it scale — crossover at $n = 4$, $36\times$ by $n = 7$





Left. The controlled evolution is the bottleneck: the standard construction builds $|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes U$ as one $(n{+}1)$-qubit `UnitaryGate`, costing $\sim 4^{\,n+1}$. Controlled AQC grows *roughly linearly*, so the crossover is structural — measured, not extrapolated. **Right.** The trap, next slide. [R046]

And on a 2D lattice? AQC-Tensor is an MPS method, native to 1D, so we checked. On a 2×4 grid the crossover moves right by *one step* ($n{=}4$ becomes a $0.92\times$ loss) and the $n{=}8$ advantage falls $122.8\times \to 100.8\times$ — an 18% cost for nearly doubling the connectivity. **It does not break.** Infidelity was in fact *better* in 2D (2.0×10^{-5}–2.0×10^{-4}); the MPS concern is deferred, not refuted, and we do not extrapolate past $n{=}8$. [R049]

The trap: a phase-blind compiler breaks a phase interferometer

`qiskit-addon-aqc-tensor` maximises **state fidelity** $|\langle\psi_{\text{target}}|\psi_{\text{ansatz}}\rangle|$ — a *magnitude*, blind to global phase. A Hadamard test measures **exactly that phase**. So a compression that converges beautifully (state infidelity 3×10^{-4}) returns

$$W|\psi\rangle = e^{i\theta}\, U|\psi\rangle, \qquad \theta \approx 2.3\text{–}3.0 \text{ rad} \quad \Longrightarrow \quad |\Delta\chi| \approx 1.3$$

on a quantity bounded by $|\chi| \leq 1$. **The error is as large as the signal.**

Why we nearly missed it. Our earlier certificate reported $|\chi_{AB}|$ — also a magnitude. A magnitude cannot detect a phase error. [R035, now closed by R046]

The fix costs one single-qubit gate. $\theta = \arg\langle U\psi|W\psi\rangle$ is free at compile time, and a $P(-\theta)$ **on the ancilla** cancels it exactly: $|\Delta\chi|$ **1.33** $\to$ **0.008**.

Scope, not buried. (i) AQC is compressed against *one* input state, so its *process* infidelity is large ($\sim$0.96) — legitimate here, since a Hadamard test's controlled gate only ever acts on that one state, and illegitimate the moment anything feeds it another. [R046] (ii) On the **2D Fermi–Hubbard** model the gate win survives (34.7$\times$ at 8 qubits) but accuracy degrades $\sim$30$\times$: $|\Delta\chi|$ after the phase fix rises to 0.021–0.046, at or above the $\sim$0.02 shot-noise floor at 2000 shots. There the compression is cheap but *not yet accurate enough to use as-is*. [R051]

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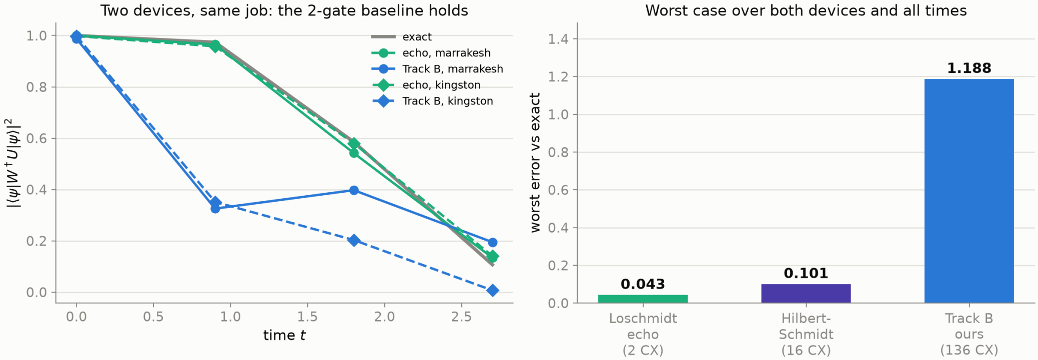
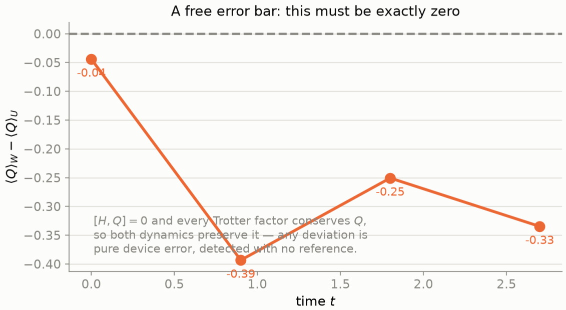
First Track B on a QPU — and the cheap baseline beat us



116 circuits × 1000 shots, all four arms in *one* job per device, so each comparison is same-device same-hour. Predictions were recorded before submission. **The 136-CX Track-B arm degrades fast** — clean at $t = 0$ (errors ≤ 0.053), noise dominated by $t = 0.9$ — and on `kingston` it goes wrong in *phase*, not just magnitude ($-0.03 - 0.59i$ against exact $+0.99 + 0.02i$ at $t = 0.9$), so this is not simple damping. The 2-CX echo holds on both devices. [R044]

The garbage also ships a free error bar



$Q = \sum_j Z_j$ commutes with H, and *every* Trotter factor conserves it — so $\langle Q\rangle_W - \langle Q\rangle_U$ is **exactly zero** for any W in this family.

Measured: -0.04, -0.39, -0.25, -0.34.

[R044]

That deviation is **pure device error, detected with no reference state, no exact diagonalisation and no simulation** — and it correctly flags the same times where the other observables went bad.

It is a diagnostic, not a correction: we do *not* post-select on it, which would bias the shadow estimators.

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What we claim, and what we do not

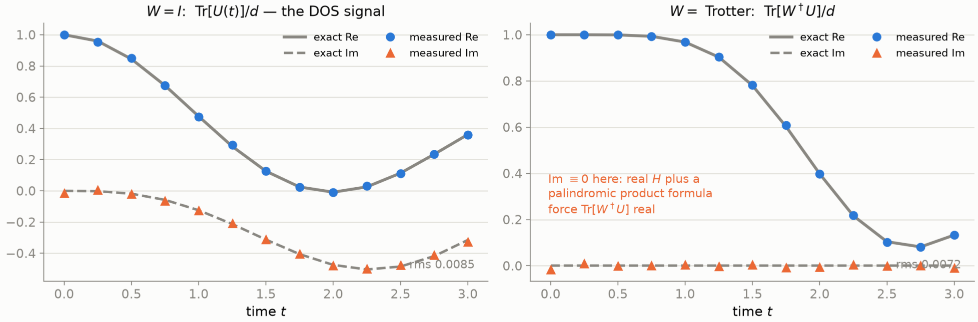
We claim

- **AQC makes the Hadamard test scale**: crossover measured at $n=4$, $36\times$ fewer two-qubit gates at $n=7$, against the honest baseline (the notebook's own direct block, not a flattering `.control()`). [R046]
- **We found and fixed a trap that silently destroys the answer** — a phase-blind objective feeding a phase interferometer — for the cost of one single-qubit gate. [R046]
- Track B **completed**: the per-observable disagreement profile via the X-basis readout, which our own earlier run left undone, validated to 4×10^{-15}. [R042]
- Track B run on a **QPU for the first time here**, against two literature baselines in the same job. [R044]

We do not claim

- **That AQC helps at small n.** Below the crossover it is *worse* — $2.6\times$ at $n=3$. And the compressed circuit is valid **only for the state it was compiled against** (process infidelity ~ 0.96). [R046]
- **That our arm is the best way to get the scalar overlap.** It is not — a 2-gate Loschmidt echo is cheaper and, on hardware, $27\times$ more accurate. [R043, R044]
- Any advantage over classical computation. These are gate counts, not runtimes; the benchmark is small *on purpose*, so every number is graded against truth.

Future work: the same circuit is a density-of-states estimator



Feed Track B the **maximally mixed** state instead of $|\psi\rangle$ and the ancilla measures $\mathrm{Tr}[W^\dagger U]/d$. With $W = I$ that is $\mathrm{Tr}[U(t)]/d$ — the generating function whose Fourier transform is the **density of states**, i.e. Eq. (7) of Goh & Koczor (arXiv:2407.03414v2). So their DOS protocol is the $W{=}I$ special case of our Track B circuit, not a separate experiment. Verified to 2.2×10^{-15}; measured on the simulator at rms 0.0085 and 0.0072. [R045]

Reproduce it

- `evidence/scripts/track_b_ab_tester.py` — Track B derivation + statevector validation + simulator study [R034]
- `evidence/scripts/track_b_baselines.py` — Loschmidt echo, Hilbert–Schmidt test, $n=2,3,4$ cost scaling [R043]
- `evidence/scripts/track_b_hardware_submit.py` — validate-then-submit, all four arms [R042]
- `evidence/scripts/track_b_hardware_fetch.py` — the analysis behind every hardware number here [R044]
- `evidence/scripts/aqc_compression.py` — AQC + Track-B certification [R035]

`github.com/mnsh0409/Qiskit-Hackathon-2026`

(full repo; this deck is `deck/trackb.pdf`)

Team 8 — Garbage Collectors

Thank you — questions welcome.

[R0xx] cites a numbered row in `RESULTS.md`

Appendix — derivations

A1 anti-control A2 the Track B identity A3 the two readouts

A4 classical shadows A5 shadows on the Track B circuit

A1 — Anti-control, and A2 — the Track B identity

A1. A controlled gate fires on $|1\rangle$. Conjugating the ancilla by X swaps the two branches:

$$X_a\Big(|0\rangle\langle 0|\otimes \mathbb{1} + |1\rangle\langle 1|\otimes W\Big)X_a \;=\; |1\rangle\langle 1|\otimes \mathbb{1} + |0\rangle\langle 0|\otimes W$$

so $X{-}cW{-}X$ applies W exactly when the ancilla is $|0\rangle$. Composing it with an ordinary cU puts a *different* evolution on each branch.

A2. Propagating $|0\rangle_a|\psi\rangle$ through the circuit:

$$\xrightarrow{H} \tfrac{1}{\sqrt{2}}(|0\rangle+|1\rangle)|\psi\rangle \xrightarrow{XcWX} \tfrac{1}{\sqrt{2}}\big(|0\rangle W|\psi\rangle+|1\rangle|\psi\rangle\big) \xrightarrow{cU} \tfrac{1}{\sqrt{2}}\big(|0\rangle W|\psi\rangle+|1\rangle U|\psi\rangle\big)$$

$$\xrightarrow{P(\phi)} \tfrac{1}{\sqrt{2}}\big(|0\rangle W|\psi\rangle+e^{i\phi}|1\rangle U|\psi\rangle\big) \xrightarrow{H} \tfrac{1}{2}\Big[|0\rangle(W+e^{i\phi}U)|\psi\rangle+|1\rangle(W-e^{i\phi}U)|\psi\rangle\Big]$$

Using $\big\|(W\pm e^{i\phi}U)\psi\big\|^2 = 2\pm 2\,\mathrm{Re}\big[e^{i\phi}\langle\psi|W^\dagger U|\psi\rangle\big]$,

$$\langle Z_a\rangle_\phi = \tfrac{1}{4}\Big(\big\|(W+e^{i\phi}U)\psi\big\|^2-\big\|(W-e^{i\phi}U)\psi\big\|^2\Big) = \mathrm{Re}\big[e^{i\phi}\langle\psi|W^\dagger U|\psi\rangle\big]$$

$W=\mathbb{1}$ returns the standard test, so this is a strict generalisation. [verified numerically to 4.56×10^{-15}, R042]

A3 — the two readouts: why deleting one gate gives the difference

Stop *before* the final Hadamard. With $|\Phi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle W|\psi\rangle + e^{i\phi}|1\rangle U|\psi\rangle\big)$ and $\rho = |\psi\rangle\langle\psi|$,

$$\rho_{\text{out}} = |\Phi\rangle\langle\Phi| = \tfrac{1}{2}\Big[|0\rangle\langle 0| \otimes W\rho W^\dagger + |1\rangle\langle 1| \otimes U\rho U^\dagger + e^{-i\phi}|0\rangle\langle 1| \otimes W\rho U^\dagger + \text{h.c.}\Big]$$

The off-diagonal ancilla blocks are killed by both partial traces below, so *both results are independent of* ϕ — one quadrature suffices:

$$\mathrm{tr}_a\big[\rho_{\text{out}}\big] = \tfrac{1}{2}\big(W\rho W^\dagger + U\rho U^\dagger\big) \equiv S, \qquad \mathrm{tr}_a\big[(Z \otimes \mathbb{1})\rho_{\text{out}}\big] = \tfrac{1}{2}\big(W\rho W^\dagger - U\rho U^\dagger\big) \equiv D$$

Hence for *any* observable O, from the same shots,

$$\langle O\rangle_W = \mathrm{Tr}\big[O(S+D)\big], \qquad \langle O\rangle_U = \mathrm{Tr}\big[O(S-D)\big]$$

Keeping the final H gives the overlap; deleting it gives the per-observable profile. [verified to 4.44×10^{-15} over 4 times $\times$ 6 observables, R042]

A4 — classical shadows: the inversion and its variance

Measure each qubit in a basis b_j drawn uniformly from $\{X, Y, Z\}$, with $V_b W V_b^\dagger = Z$. The single-qubit measurement channel is

$$\mathcal{M}_1(\sigma) = \mathop{\mathbb{E}}_{b,s}\Big[\langle s|V_b \sigma V_b^\dagger|s\rangle \; V_b^\dagger|s\rangle\langle s|V_b\Big] = \tfrac{1}{3}\sigma + \tfrac{1}{3}\mathrm{Tr}(\sigma)\nVdash$$

— expand σ over $\{\nVdash, X, Y, Z\}$: the identity part always survives, and each traceless part survives only when *its own* basis is drawn, with probability $1/3$. Inverting on the traceless part,

$$\mathcal{M}_1^{-1}(A) = 3A - \mathrm{Tr}(A)\nVdash \quad \Longrightarrow \quad \hat{\rho} = \bigotimes_j \Big(3\, V_{b_j}^\dagger |s_j\rangle\langle s_j| V_{b_j} - \nVdash\Big), \qquad \mathbb{E}[\hat{\rho}] = \rho$$

For a Pauli string P, the single-qubit traces are 1 ($P_j = \nVdash$), $3s_j$ ($P_j = b_j$), and 0 otherwise, so

$$\hat{P} = 3^{\,w(P)} \prod_{j \in \mathrm{supp}(P)} s_j \,\mathbf{1}\big[b_j = P_j\big], \qquad \mathbb{E}\big[\hat{P}^2\big] = 3^{2w} \cdot \underbrace{3^{-w}}_{\text{all } w \text{ match}} = 3^{\,w(P)}$$

The second moment is **exactly** 3^w, independent of ρ — which is why low-weight observables are cheap and weight-n ones are not. [HKP, *Nat. Phys.* **16**, 1050 (2020); CONVENTIONS §(Pauli-string estimator)]

A5 — putting them together on the Track B circuit

Each shot of arm B records $(b_0 \ldots b_{n-1},\, s_0 \ldots s_{n-1},\, a)$ with $a = \pm 1$ the ancilla outcome. Form $\hat{P}$ from the system record as in A4. Then, by A3,

$$\mathbb{E}\big[\hat{P}\big] = \mathrm{Tr}[P\,S] = \tfrac{1}{2}\big(\langle P\rangle_W + \langle P\rangle_U\big), \qquad \mathbb{E}\big[a\,\hat{P}\big] = \mathrm{Tr}[P\,D] = \tfrac{1}{2}\big(\langle P\rangle_W - \langle P\rangle_U\big)$$

so a single sample mean of each gives both dynamics separately, for every P at once:

$$\widehat{\langle P\rangle_W} = \overline{\hat{P}} + \overline{a\hat{P}}, \qquad \widehat{\langle P\rangle_U} = \overline{\hat{P}} - \overline{a\hat{P}}$$

Two consequences we use on the slides.

- Any $O = \sum_P c_P P$ follows by linearity — H, Q, and the hopping term all come from the same records.
- If $[O, H] = 0$ and every factor of W conserves O, then $\langle O\rangle_W - \langle O\rangle_U \equiv 0$ *identically*. Its measured value is therefore **pure device error**, with no reference needed — the free error bar on slide 19. [R044]

A6 — Track B $\to$ density of states, and where the phase matters

With $\rho = \mathbb{1}/d$ the Track B identity of A2 becomes state-*independent*:

$$\chi_{AB}(t) = \mathrm{Tr}\left[W^\dagger U \rho\right] \xrightarrow{\rho=\mathbb{1}/d} \frac{1}{d}\mathrm{Tr}\left[W^\dagger U\right] \qquad \text{and at } W = \mathbb{1}: \quad \frac{1}{d}\mathrm{Tr}[U(t)]$$

which is Eq. (7) of Goh & Koczor up to normalisation. Since $\mathrm{Tr}[A] = \sum_z \langle z|A|z\rangle$, the mixed input is realised by enumerating the d computational basis states — the deterministic version of their Statement 2 (a 1-design, e.g. random $\{\mathbb{1}, X\}^{\otimes n}$, suffices).

Where the phase is worth having — measured, not assumed:

	max $\lvert\mathrm{Im}\rvert$	does the HST lose information?
$W = \mathbb{1}$ (the DOS signal)	**0.504**	**yes** — it returns magnitude only
W = 2nd-order Suzuki (Trotter error)	2.8×10^{-14}	**no** — it is real anyway

Real H gives $U^* = U^\dagger$, and a *palindromic* product formula gives $W^* = W^\dagger$, which forces $\mathrm{Tr}[W^\dagger U]$ real. So our advantage over the HST is the full complex value *for the DOS case*, and only the register size ($n+1$ vs $2n$) for the Trotter case. **We checked this rather than claiming the phase everywhere.** A measured nonzero Im in the second row is therefore another reference-free error signal. [R045]